

PSYCH 315H Honors Cognitive Psychology / PSYCH 615 Cognitive Psychology & Neuroscience

MWF 11:15a - 12:05p

Tobin Hall 307

Instructor: Dr. Lauren Aulet

Office Hours: Tuesdays at 3-4pm, or by appointment (Tobin 425)

Summary: Cognitive psychology is the branch of psychology concerned with the scientific study of the mental processes involved in perception, attention, memory, knowledge, language, problem solving, reasoning, and decision-making. In short, cognitive psychology is concerned with the scientific study of the mind and mental processes. This is an intensive course that will cover the different aspects of Cognitive Psychology in some depth at a level appropriate for advanced undergraduates and beginning graduate students. You will be learning from a textbook and primary journal articles. In addition to exams, students will give an in-class presentation.

Readings: The textbook for this course is *Cognition* by Daniel Reisberg. It is available from eCampus, and from other sources. Be sure that you are using the 8th edition. You can use one of the printed versions (hardcover, paperback, loose-leaf) or the electronic version. We will not be using the ZAPS experimental package that comes with some versions of the textbook, so it is fine to buy or rent a used copy without this package.

Date	Topic	Reading	Presentation
9/3	Syllabus & What is Cognitive Psychology?	Chapter 1: The Science of the Mind	
9/5	History of Cognitive Psychology	Chapter 1: The Science of the Mind	Presentation sign up
9/8	Cognitive Psychology - Behavioral	Moyer & Landauer 1967	Presentation 1: Moyer & Landauer 1967
9/10	Neuroanatomy	Chapter 2: The Neural Basis for Cognition	
9/12	Neural Methods	Pitcher 2009	Presentation 2: Pitcher 2009
9/15	Low level vision	Chapter 3: Visual Perception	
9/17	Low level vision	Chapter 3: Visual Perception	
9/19	High level vision	Goodale et al., 1991	Presentation 3: Goodale et al., 1991
9/22	High level vision	Chapter 3: Visual Perception	
9/24	Object perception	Chapter 4: Recognizing objects	
9/26	Face perception	Chapter 4: Recognizing objects	
9/29	Word perception	Chapter 4: Recognizing objects	
10/1	Exam 1		

10/3	Visual Imagery	Chapter 11: Visual Knowledge Shepard & Metzler 1971	Presentation 4: Shepard & Metzler 1971
10/6	Visual Imagery	Chapter 11: Visual Knowledge Chang et al 2025	Presentation 5: Chang et al 2025
10/8	Number	Halberda et al 2006	Presentation 6: Halberda et al 2008
10/10	Number and Space	Zorzi et al 2002	Presentation 7: Zorzi et al 2002
10/13	No class (Indigenous people's day)	Chapter 5: Paying Attention	
10/15	Attention	Chapter 5: Paying Attention	
10/17	Attention / Working Memory	Chapter 6: The Acquisition of Memories and the Working-Memory System	
10/20	Working Memory	Chapter 6: The Acquisition of Memories and the Working-Memory System	
10/22	Long term Memory	Chapter 7: Interconnections between Acquisition and Retrieval Brady et al., 2008 PNAS	Presentation 8: Brady et al., 2008 PNAS
10/24	Retrieval	Chapter 7: Interconnections between Acquisition and Retrieval	
10/27	Concepts - behavior	Chapter 9: Concepts and Generic Knowledge	
10/29	Concepts - brain	Chapter 9: Concepts and Generic Knowledge	
10/31	Exam 2		
11/3	Lauren Out – No class		
11/5	Language - Comprehension	Chapter 10: Language	Presentation 9: Saffran, Aslin, & Newport, 1996
11/7	Language - Deficits, Brain	Chapter 10: Language	
11/10	Language and thought	Winawer et al., 2007 PNAS	Presentation 10 – Winawer et al., 2007 PNAS
11/12	Analogy	Smirnova et al., 2015	Presentation 11 – Smirnova et al., 2015
11/14	History of Intelligence Testing	Chapter 13: Problem Solving and Intelligence	
11/17	Intelligence	Chapter 13: Problem Solving and Intelligence	
11/19	Intelligence	Chapter 13: Problem Solving and Intelligence	

11/21	Judgment & Reasoning	Gazes et al., 2017, Dev Sci	Presentation 12: Gazes et al., 2017, Dev Sci
11/24	Judgment & Reasoning	Chapter 12: Judgment & Reasoning	
12/1	Consciousness	Chapter 14: Conscious Thought, Unconscious Thought	
12/3	Consciousness	Chapter 14: Conscious Thought, Unconscious Thought	
12/5	Exam 3		
12/8	Final review		
12/17	Final exam	***At 10:30 AM	

Grading / Assignments:

Exam 1: 100 points

Exam 2: 100 points

Exam 3: 100 points

In-class research presentation: 50 points

Final exam (cumulative): 100

Lowest exam grade will be dropped. Final grade out of 350 points

Exams will contain: 15 multiple choice (4 points each) [60]

4 10 point short response [40]

No extra credit via SONA systems is offered for this course.

Grade Scale:

A	93-100	D+	67-69
A-	90-92	D	63-66
B+	87-89	F	0-62
B	83-86		
B-	80-82		
C+	77-79		
C	73-76		
C-	70-72		

Academic Honesty:

Since UMass Amherst is strongly committed to academic integrity, which is defined as completing all academic work without cheating, lying, stealing, or receiving unauthorized assistance from any other person, or using any source of information not appropriately authorized or attributed. As a community, we hold each other accountable and support each other's knowledge and understanding of academic integrity. Academic dishonesty is prohibited in all programs of the University and includes, but is not limited to: Cheating, fabrication, plagiarism, lying, facilitating dishonesty, via analogue and digital means. Sanctions may be imposed on any student who committed or participated in an academic integrity infraction. Any person who has reason to believe that a student has committed an academic integrity infraction should bring such information to the attention of the appropriate course instructor as soon as possible. All students at the University of Massachusetts Amherst have read and acknowledged the Commitment to Academic Integrity and are knowingly responsible for completing all work with integrity and in accordance with the policy: <https://www.umass.edu/senate/book/academic-regulations-academic-integrity-policy>

Accommodation:

The University of Massachusetts Amherst is committed to providing an equal educational opportunity for all students. If you have a documented physical, psychological, or learning disability on file with Disability Services (DS), you may be eligible for reasonable academic accommodations to help you succeed in this course. If you have a documented disability that requires an accommodation, please notify me within the first two weeks of the semester so that we may make appropriate arrangements. For further information, please visit Disability Services (<https://www.umass.edu/disability/>)

Diversity:

In recognition and affirmation of the worth and dignity of all persons, the instructor of this class is dedicated to upholding the values of diversity and inclusion and to minimizing disparities in access to learning. All students who are willing to learn are equally welcome without regard to age, race, ethnicity, birthplace, sex, gender identity, disability, sexual orientation or identity, socioeconomic status, political party or beliefs, or other group affiliation or personal characteristics. All students are likewise expected to accord each other the same level of respect, bearing in mind that becoming more respectful and inclusive is a journey rather than a destination. We will all have stumbling points along the way, including the instructor, and these are often valuable learning opportunities. If I ever say or do anything insensitive in this class, I sincerely welcome students to call me out or communicate their discomfort to me in person after class, during office hours, by email, or anonymously.

Title IX:

In accordance with Title IX of the Education Amendments of 1972 that prohibits gender-based discrimination in educational settings that receive federal funds, the University of Massachusetts Amherst is committed to providing a safe learning environment for all students, free from all forms of discrimination, including sexual assault, sexual harassment, domestic violence, dating violence, stalking, and retaliation. This includes interactions in person or online through digital platforms and social media. Title IX also protects against discrimination on the basis of pregnancy, childbirth, false pregnancy, miscarriage, abortion, or related conditions, including recovery. There are resources here on campus to support you. A summary of the available Title IX resources (confidential and non-confidential) can be found at the following link: <https://www.umass.edu/titleix/resources>. You do not need to make a formal report to access them. If you need immediate support, you are not alone. Free and confidential support is available 24 hours a day / 7 days a week / 365 days a year at the SASA Hotline 413-545-0800.